Majid Farhadi

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Education

University of Alberta, Computer Engineering — Edmonton, Canada, 2023 – 2026

Isfahan University of Technology, Computer Engineering — Isfahan, Iran, 2018 – 2023

Experience

Very Large Scale Integration (VLSI) Research Lab, University of Alberta, Graduate Research Assistant — Edmonton, Canada, 2023 – 2026, 3 years

- Designed and implemented AI architectures, including training and inference circuits, on FPGAs.
- Integrated and synchronized IP core modules with edge detector 1-stage synchronizers.
- Wrote testbench modules for behavioral, functional, and timing simulation and verification of RTL designs.
- Defined implementation constraints including clock period to meet timing requirements.
- Optimized dynamic power by minimizing switching activity through stop conditions.
- Fine-tuned distance functions in clustering algorithms and derived gradients in backpropagation algorithms.

Structured Machine Learning Research Lab, Simon Fraser University, Graduate Research Assistant — Burnaby, Canada, 2023 – 2024, 1 year

- Counted motifs in a graph database by matrix multiplication, suitable for ML applications.

Embedded Systems and Networks Research Lab, Isfahan University of Technology, Undergraduate Research Assistant — Isfahan, Iran, 2021 – 2023, 2 years

- Designed an interface for a hardware accelerator for the time-slotted channel hopping (TSCH) protocol.